

Do democracies actually work?

If a majority of citizens want a policy, does it become law?

Most people assume “yes, eventually.” That’s the whole pitch of representative democracy: preferences in, policy out, some friction in the middle.

I wanted to know how much friction. So I built an agent-based model of four democratic legislatures (pure parliamentary, pure presidential, and the two semi-presidential hybrids) and ran each one 200 times across four political scenarios: baseline, fragmented, polarised, small.

That’s 3,200 simulated legislative sessions. The code is open source at <https://github.com/tofuadmiral/institutional-representation-abm> and there’s a live demo at <https://institutional-representation-abm.streamlit.app> where you can move every dial yourself.

Three things surprised me.

1. Parliaments don’t fail gracefully. They collapse.

Under fragmentation (five small parties, no natural majority) a pure parliamentary system passes **0.05%** of bills.

Not fewer bills. Not slower bills. Effectively zero.

My first explanation was coalition math: no majority forms, so nothing passes. It matched decades of arguments about Italy and Israel, and it survived 200 seeds with bootstrap confidence intervals (which was a relief, nothing worse than a headline finding that dies at $n=200$). It was also the wrong explanation, which brings me to the fun part.

2. The thing that kills parliaments also saves them

A sharp early reader kept poking at my assumptions and asked what MPs in the model actually do when no government forms. Good question. The answer was buried in the code: with no coalition, every single MP got treated as opposition and whipped to vote no. All 24 of them, against everything, forever.

That’s an assumption doing a lot of silent work. So I promoted it to a setting with two positions and measured both:

- **cohesive obstruction:** a hung parliament behaves like Weimar. Blocs whip against all business regardless of content.
- **personal vote:** no government means no whip. Every MP votes their own preference (the minority-governance world Kaare Strøm documented back in the 80s: roughly a third of postwar European cabinets were minority governments, and they governed fine).

Same fragmented chamber, both settings, 200 seeds each:

Institution	Cohesive obstruction	Personal vote
Parliamentary	0.05%	46.4%
Premier-presidential	1.3%	33.3%
President-parliamentary	8.6%	8.6%
Republican	44.8%	44.8%

Read that top row again. A fragmented parliament where MPs vote their own preferences passes bills at the same rate as a presidential system (46.4% vs 44.8%, overlapping confidence intervals). Failing to form a government blocks nothing by itself. The collapse requires a cohesive opposition whipping against everything. The pathology lives in the whip, specifically the opposition’s copy of it.

An independent check lands on the same answer: disable party discipline entirely and fragmented parliamentary passage comes out at 46.7%. Same number, different door.

And the whip cuts the other way under polarisation (two ideologically distant parties, wide policy proposals): there, discipline is what keeps passage alive, and removing it costs the parliament 66 percentage points. The size of the effect scales cleanly with how much each institution depends on forming a majority government. Under fragmentation the discipline ablation moves parliamentary by +0.47, premier-presidential by +0.31, president-parliamentary by +0.23, and republican by −0.24.

So discipline aggregates legislators into blocs, and everything depends on which bloc it builds. A governing coalition near the median passes everything. An opposition with nothing to govern blocks everything. Most reform debates quietly assume one of those worlds and prescribe for it. The model says check which world you’re in first.

3. There is no best system, just a dial

I added a metric called the **policy representation gap**: the L2 distance between the bills that pass and the median preference of the constituencies they’re supposed to represent.

Under polarisation:

Institution	Passage rate	Representation gap
Parliamentary	88.2%	+0.037
Premier-presidential	72.5%	−0.012
President-parliamentary	73.0%	−0.010
Republican	21.8%	− 0.530

The parliamentary system passes four times as many bills, but they drift away from what voters want. The presidential system passes far fewer, and the ones that survive the veto sit much closer to the public’s median.

This is the deepest finding in the project. Throughput and fidelity live on the same spectrum. You cannot max both. Every real democracy is a point on this curve, and “more democratic” turns out to be a choice about which axis you’re willing to give up.

So, do democracies work?

Depends what “work” means.

- *Passes what the majority wants, fast*: parliaments, right up until the opposition organizes into a wall.
- *Only passes laws that match the public*: presidential systems, at the cost of long stretches where nothing passes at all.
- *Survives whichever crisis shows up*: the hybrids are doing something interesting in the middle. They lose on both metrics individually but lose less when the scenario shifts. There’s a real case for designs whose worst case is mediocre rather than catastrophic.

One caveat: the model is generative, in the Kollman-Miller-Page and Laver-Sergenti tradition. It isolates structural tradeoffs, it doesn’t forecast what Italy does next year. But the tradeoffs it isolates are sharp, reproducible (every result is seed-deterministic and CI-gated), and uncomfortable for anyone whose theory of democracy has only one number in it.

Model, paper, sensitivity analyses, and the interactive demo are all here: <https://github.com/tofuadmiral/institutional-representation-abm>. Go break a parliament, it’s weirdly fun.